

Chapter 12 — Fundamentals of Intraoperative Patient Safety — Numbers, Gold Standards and Traps

IF YOU READ NOTHING ELSE

- **Universal Protocol** (Joint Commission 2004): verification → site marking → time-out
- **WHO (World Health Organization) checklist 2008**: 19 items; death ↓ 47%, complications ↓ 36%
- **Count discrepancy** → ~100× odds of retained surgical item (RSI)
- **Commonest RSI**: sponge (~2/3); then drains (Penrose); then needles
- **RSI definition**: found after skin closure, even still anaesthetised

- **Mark correct site only**; never "No"/"X"
- **Fire triangle**: oxidiser (anaesthesia), fuel (nursing), ignition (surgeon)
- **Airway surgery**: O₂ ≤ 30%, stop nitrous oxide (N₂O), 3–5 min before diathermy
- **Alcohol prep**: dry ≥ 3 min; pooling ↑ fire risk
- **Communication failure**: up to 70% of OR (operating room) adverse events

BACKGROUND AND CHECKLISTS

- **To Err Is Human (IOM, Institute of Medicine, 1999)**: 44,000–98,000 deaths/year
- **2016 estimate**: medical error third leading US cause of death
- **Adverse events**: ~50% surgical; ~50% of those preventable
- **Malpractice errors**: ~75% intraoperative
- **Pronovost 2006**: central-line checklist, 108 intensive care units → bloodstream infection ↓ 66%
- **WHO pilot**: 8 institutions, ~4000 operations
- **WHO phases**: sign in (before induction) → time out (before incision) → sign out (before leaving)
- **Sign-in blood-loss flag**: >500 mL (7 mL/kg children)
- **Time-out antibiotic check**: given within last 60 min

WRONG-SITE SURGERY PREVENTION

- **Incidence**: ~1 in 112,000 — sentinel never event
- **Definition**: wrong side, wrong patient, wrong part
- **Joint Commission**: founded 1951; accredits >20,000 organisations
- **Up to 30% of root causes**: before the day of surgery
- **Identifiers**: full name, date of birth, record number — barcoded band
- **Consent**: list contingent procedures
- **Interpreters**: professional, never family
- **Marker**: present at time-out; patient awake
- **Lower-tract endourology**: mark wristband, not meatus
- **Repeat time-out**: surgeon change or new procedure
- **Time-out errors**: music on, member busy/on break, "we all know the patient"

SYSTEMS FAILURES

- **Swiss cheese model**: harm when holes align; stopped = near miss
- **Latent** (organisation, supervision, preconditions) vs **active** (unsafe acts) failures
- **Wrong-site risk**: emergency, deformity/obesity, time pressure, multiple surgeons/procedures
- **Checklists in urgent cases**: ↓ complications and mortality; no start delay
- **Root causes**: hierarchy, staffing shortage, temporary staff, pandemics

RETAINED SURGICAL ITEMS

- **Incidence**: 1 in 10,000 to 1 in 2800
- **Risk**: emergency, change of plan, obesity, multiple teams, open cavity, many items, wrong count, long case, attending absent at closure

- **Trainee present**: lower risk
- **Human factors** (leadership, communication): dominant cause
- **Count standard**: AORN (Association of periOperative Registered Nurses)
- **Discrepancy before closure** → delay closure, find item
- **Routine imaging**: most RSIs found had correct counts
- **Barcoded sponges count**; **RF (radiofrequency) tags** detect — accurate even in obesity

SHARPS SAFETY

- **Universal Precautions**: late 1980s, after human immunodeficiency virus epidemic
- **Blood contact**: up to half of operations
- **Surgeons + first assistants**: ~60% of injuries; ≤24% by colleague
- **1/3 of injuring sharps** recontact the patient
- **Single-glove puncture**: up to 61%, ~40 min in
- **Double gloving**: blood on solid needle ↓ 95%
- **Blunt needles**: fascia and muscle closure
- **Designated sharps zone**: ACS (American College of Surgeons) best practice; no hand-to-hand

FIRE SAFETY

- **US incidence**: ~600/year
- **Site**: 85% head, neck, upper chest; 81% under monitored anaesthesia care
- **Oxidiser**: O₂ in 95%; N₂O equals O₂ as oxidiser
- **Ignition**: monopolar "Bovie" ~90%; bipolar lowers, not abolishes
- **ChlorPrep**: ~70% isopropyl alcohol; povidone-iodine/aqueous chlorhexidine non-flammable
- **Laser**: low risk under irrigation; cracked fibre → fire
- **Silverstein score** (1 point each): open O₂, ignition source, above xiphoid → 1 low, 2 intermediate, 3 high
- **Most urology**: low/intermediate
- **Sequence**: pause → saline/water → drapes to floor → extinguisher → RACE
- **RACE**: Rescue, Alert, Confine (gases off, doors shut), Extinguish/evacuate

DISTRACTIONS AND COMMUNICATION

- **Sterile cockpit**: silence in pre-defined critical steps (e.g. hilar clamping)
- **ACS smartphones**: silence, no personal calls, redirect, distinct urgent ringtone
- **Alarm fatigue** → individualise alarm settings

- **Communication model:** sender, receiver, message, channel; encode/decode; noise
- **OR communication failure:** ~30% of exchanges; ⅓ create risk
- **Communication risk:** cognitive load, bias, hierarchy, diffusion of responsibility, handoffs
- **Situational awareness:** where from, where now, where going
- **TeamSTEPPS:** AHRQ (Agency for Healthcare Research and Quality) + Department of Defense, 2006
- **Five principles:** structure, communication, leadership, situation monitoring, mutual support
- **SBAR:** situation, background, assessment, recommendation
- **Leadership tools:** brief, huddle, debrief
- **STEP:** status of patient, team, environment, progress
- **I'M SAFE:** illness, medication, stress, alcohol/drugs, fatigue, eating/elimination
- **Two-challenge rule:** voice concern at least twice
- **CUS:** Concerned, Uncomfortable, Safety issue — stops the line
- **Simulation (in situ/high-fidelity)** > didactic alone

CLASSIC TRAPS

- **Trap:** correct count excludes RSI — **Actually:** most found RSIs had correct counts
- **Trap:** RSI only if found after leaving theatre — **Actually:** after skin closure
- **Trap:** mark non-operative side "No" — **Actually:** mark correct site only
- **Trap:** family can interpret consent — **Actually:** professional interpreter
- **Trap:** skip checklist in emergency — **Actually:** most needed; ↓ mortality
- **Trap:** N₂O safe oxidiser — **Actually:** equals O₂
- **Trap:** 3 min drying abolishes fire — **Actually:** fires still occur, worse with pooling
- **Trap:** MAC here = alveolar concentration — **Actually:** monitored anaesthesia care
- **Trap:** trainee presence raises RSI risk — **Actually:** lowers it
- **Trap:** didactic TeamSTEPPS improves outcomes — **Actually:** no better than none